E-Voting System

by Group-I

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Outline

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Introduction

- A web-based voting system
- Enables elections digitally
- A secure and efficient process
- Replaces traditional and physical voting
- Saves both time and cost

Motivation

- Traditional voting systems are **time-consuming and costly**
- Difficult to manage elections in **remote areas of Nepal**
- Limited accessibility for voters (distance, mobility, abroad)
- Growing need for **digital and efficient solutions**
- Motivation to build a **simple academic model of e-voting**

Problem Statement

- Traditional Voting is slow
- High cost in resources
- Prone to human mistakes
- Risk of illegal voting
- Lack of transparent voting

Literature Review

Electronic voting systems have been widely studied as an alternative to traditional voting methods

Research highlights key benefits such as:

- Faster result generation
- Improved accuracy
- Reduced manual effort

However, many proposed systems:

- Focus on high-level security techniques
- Require complex infrastructure
- Are difficult to implement in real-world conditions

Contd. Literature Review

Studies related to Nepal indicate challenges like:

- Geographic barriers in rural areas
- Limited access to reliable internet
- Need for cost-effective solutions

Objectives

- Develop a secure and proper voting system
- Ensuring fair elections
- Prevention of duplicate and illegal voting
- Providing fast results
- Transparent voting

Key features

- Secure login and authentication
- Vote counting after the election
- One voter one vote
- Transparent elections
- Remote Accesibility
- Admin control

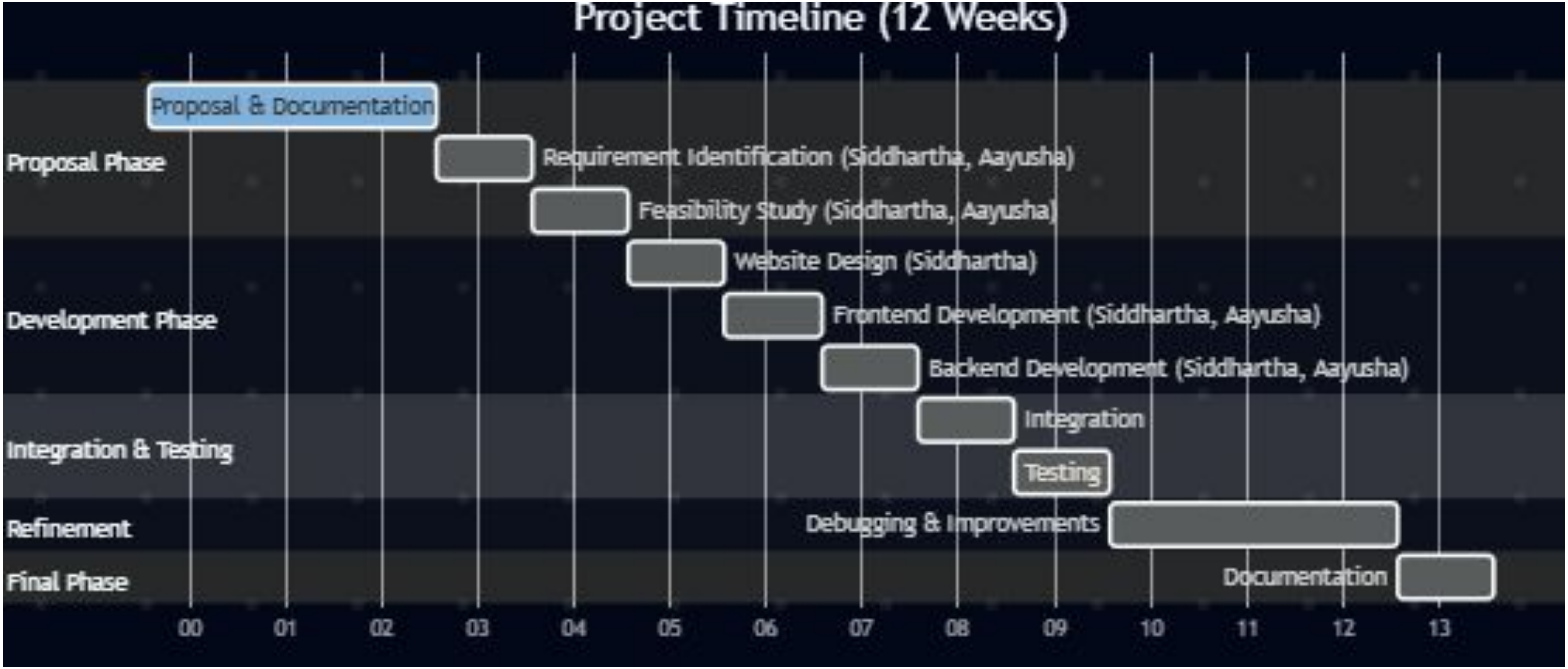
Methodology

- Requirement Identification
- Feasibility Study
- High-Level Design
- Development
- Testing
- Deployment

Contd of methodology

- Study of existing voting systems
- Identification of user needs
- Design of a system flow
- Development of frontend and backend
- testing system security
- Deployment on server

Gantt chart



Project Timeline (12 Weeks)
Proposal & Documentation
Proposal Phase
Requirement Identification (Siddhartha, Aayusha)
Feasibility Study (Siddhartha, Aayusha)
Website Design (Siddhartha)
Development Phase
Frontend Development (Siddhartha, Aayusha)
Backend Development (Siddhartha, Aayusha)
Integration
Integration & Testing
Testing
Refinement
Debugging & Improvements
Final Phase
Documentation
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Tech Stack

- Frontend: HTML, CSS, JavaScript, Minimal React
- Backend: PHP
- Database: MySQL
- Tools & Environment:XAMPP (Local server)
- VS Code (Development)
- Version Control: Git

Feasibility

- **Technical**: Using tools such as PHP, MySQL, React, JS, HTML, Css, Postman
- **Operational:** Easy and practical to use. Tools: Figma, Canva
- **Economic:** A low cost system

Comparison between existing systems in Nepal

Current System(Nepal):

- Managed by Election Commission of Nepal
- Uses Electronic Voting Machines (EVMs) and manual processes
- Physical presence required

Limitations:

- Time-consuming vote counting
- Logistical challenges in remote areas (Himalayan & rural regions)
- High cost of organizing elections
- Limited accessibility for citizens living far from the polling stations.

Expected Outcomes

- Secure login and authentication system
- One user can vote only once
- Automated vote counting
- User-friendly interface

Conclusion

- Proposes a web-based e-voting system for national elections in Nepal
- Aims to improve accessibility, efficiency, and result speed
- Addresses challenges in the current system managed by the Election Commission of Nepal
- Highlights potential for remote voting and automated counting

Thank you